

Cocoa Price Forecasting: Weighted Local Linear vs Machine Learning Methods

Project: Empirical Validation of the CGS Weighted Local Linear Framework

Reference: CGS (2025) - Nonparametric Time Series Forecasting with Structural Breaks

Abstract

This notebook implements the experimental framework to validate the CGS weighted local linear methodology on cocoa price data.

Models Evaluated

Category	Model	Data Usage	Description
NP Benchmarks	Pre-Break LL	Pre-break only	High bias, low variance baseline
	Post-Break LL	Post-break only	Low bias, high variance baseline
	Weighted LL (WLL)	Combined	CGS convex combination with MFV-tuned gamma
ML Competitors	Random Forest	Full sample	Ensemble bagging method
	XGBoost	Full sample	Gradient boosting method
ML Extension	RF Weighted Combo	Combined	CGS weighting applied to RF
	XGB Weighted Combo	Combined	CGS weighting applied to XGB

Evaluation Metrics

- **Primary:** Out-of-Sample Mean Squared Forecast Error (MSFE)
- **Secondary:** Modified Diebold-Mariano (MDM) test for pairwise comparisons

1. Environment Setup

Dependencies installed successfully.

Standard libraries imported.

Project modules imported.

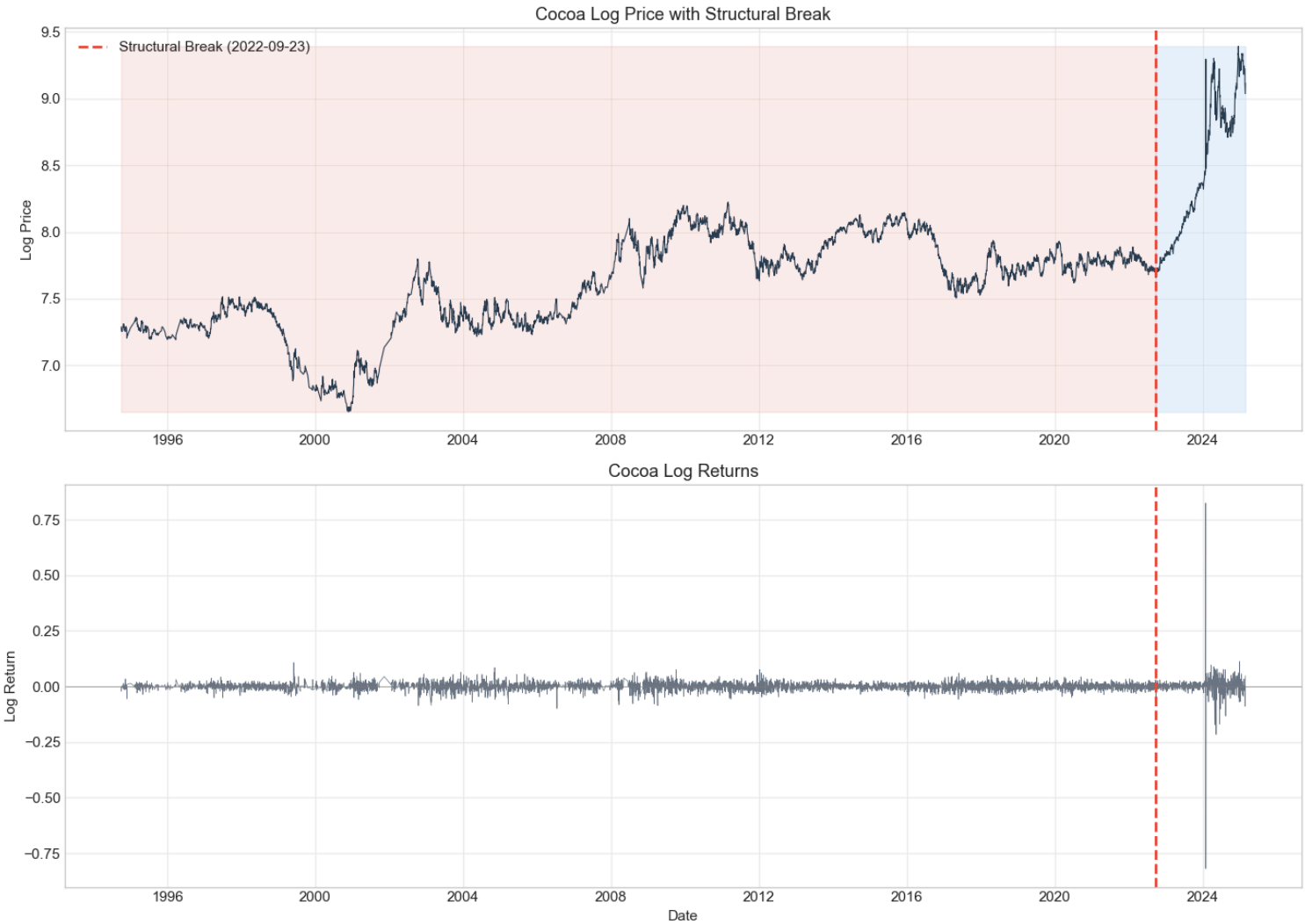
2. Data Loading and Exploration

Dataset: 6,734 observations
Date range: 1994-10-04 to 2025-02-26
Columns: ['date', 'PRCP_anom_mean', 'TAVG_anom_mean', 'PRCP_anom_std', 'TAVG_anom_std', 'N_stations', 'log_price', 'log_price_lagt', 'log_return', 'log_return_forecast_target']

	date	PRCP_anom_mean	TAVG_anom_mean	PRCP_anom_std	TAVG_anom_std	N_stations	log_price	log_price_
0	1994-10-04	0.0000	0.25	0.000000	0.000000	1	7.287752	7.290
1	1994-10-05	0.0925	0.00	0.137886	2.828427	2	7.265828	7.287
2	1994-10-06	0.0040	-0.40	0.005657	0.565685	2	7.265653	7.265
3	1994-10-07	-0.0284	0.46	0.377151	1.306905	5	7.255916	7.265
4	1994-10-10	-0.4325	-4.00	0.000000	0.000000	1	7.253987	7.255



Structural Break Configuration:
Break Index: 6116
Break Date: 2022-09-23
Pre-break observations: 6,116
Post-break observations: 618



Summary Statistics by Regime:
Pre-break - Mean: 0.000075, Std: 0.015553, N: 6116
Post-break - Mean: 0.002491, Std: 0.071729, N: 618

Methodology: Mohr-Selk (2020)

We use the Mohr-Selk (2020) detector to locate the structural break. Steps are simple: (1) pick a pilot bandwidth with MFV for a local linear fit, (2) fit the pilot mean and take residuals, (3) scan candidate break points over a trimmed

sample and choose the one with the largest KS-type statistic. The series is treated as strong mixing within each regime. In practice here, the detector lands on break index 6117 (2022-09-23), matching the configured break date.

Running Mohr-Selk (2020) Break Detection...

=====

Sample size for break detection: $T = 6694$

Dimension: $d = 1$

Step 1: Selecting pilot bandwidth via MFV cross-validation...

LocalPolynomialEngine will use device: cuda

Creating bandwidth grid with $T=6694$, $d=1$

Generated bandwidth grid: [0.01717366 0.02864739 0.04778673 0.07971307 0.13296941 0.22180635
0.36999529 0.61718934 1.02953386 1.71736599]

Set block size to 1338 for training length 6694 and $Q=4$.

Selected pilot bandwidth: $h = 1.7174$

Pilot MFV MSE: 0.000813

Step 2: Computing pilot estimates...

Pilot estimates computed for 6694 observations

Step 3: Estimating break date via KS functional maximization...

Results:

Estimated break index (1-based): 6117

Estimated break date: 2022-09-23

Configured break index: 6117

Configured break date: 2022-09-23

LocalPolynomialEngine will use device: cuda
Instantiated runner for NP_LL_Combo. Results will not be saved.
Structural break date identified: 2022-09-20
Full train/CV size: 6694, OOS test size: 40

--- (1/3) Tuning bandwidth for PRE-break model ---

Creating bandwidth grid with T=6113, d=1
Generated bandwidth grid: [0.01748836 0.02917235 0.0486624 0.08117378 0.13540603 0.22587087
0.37677533 0.62849912 1.04839973 1.74883615]
Set block size to 1222 for training length 6113 and Q=4.
Best bandwidth for PRE-break model: 1.7488

--- (2/3) Tuning bandwidth for POST-break model ---

Creating bandwidth grid with T=581, d=1
Generated bandwidth grid: [0.02800043 0.04670754 0.07791287 0.1299665 0.2167972 0.36163952
0.60325108 1.00628345 1.67858196 2.80004347]
Set block size to 116 for training length 581 and Q=4.
Best bandwidth for POST-break model: 2.8000

--- (3/3) Tuning gamma for Convex Combination model ---

Set block size to 116 for training length 581 and Q=4.

Gamma: 0.00, MFV MSE: 0.006723

Gamma: 0.05, MFV MSE: 0.006722

Gamma: 0.10, MFV MSE: 0.006722

Gamma: 0.15, MFV MSE: 0.006722

Gamma: 0.20, MFV MSE: 0.006721

Gamma: 0.25, MFV MSE: 0.006722

Gamma: 0.30, MFV MSE: 0.006722

Gamma: 0.35, MFV MSE: 0.006722

Gamma: 0.40, MFV MSE: 0.006723

Gamma: 0.45, MFV MSE: 0.006723

Gamma: 0.50, MFV MSE: 0.006724

Gamma: 0.55, MFV MSE: 0.006725

Gamma: 0.60, MFV MSE: 0.006727

Gamma: 0.65, MFV MSE: 0.006728

Gamma: 0.70, MFV MSE: 0.006730

Gamma: 0.75, MFV MSE: 0.006731

Gamma: 0.80, MFV MSE: 0.006733

Gamma: 0.85, MFV MSE: 0.006735

Gamma: 0.90, MFV MSE: 0.006737

Gamma: 0.95, MFV MSE: 0.006740

Gamma: 1.00, MFV MSE: 0.006742

Best gamma: 0.20 (MFV MSE: 0.006721)

--- Skipping Bias-Variance Decomposition ---

Successfully completed run for NP_LL_Combo (without saving artifacts).

LocalPolynomialEngine will use device: cuda

Instantiated runner for NP_LL_Combo. Results will not be saved.

Structural break date identified: 2022-09-21

Full train/CV size: 6694, OOS test size: 40

--- (1/3) Tuning bandwidth for PRE-break model ---

Creating bandwidth grid with T=6114, d=1
Generated bandwidth grid: [0.01748779 0.02917139 0.04866081 0.08117113 0.1354016 0.22586348
0.376763 0.62847856 1.04836543 1.74877893]
Set block size to 1222 for training length 6114 and Q=4.
Best bandwidth for PRE-break model: 1.7488

--- (2/3) Tuning bandwidth for POST-break model ---

Creating bandwidth grid with T=580, d=1
Generated bandwidth grid: [0.02801008 0.04672364 0.07793972 0.13001129 0.2168719 0.36176414
0.60345895 1.0066302 1.67916038 2.80100834]
Set block size to 116 for training length 580 and Q=4.
Best bandwidth for POST-break model: 2.8010

--- (3/3) Tuning gamma for Convex Combination model ---
Set block size to 116 for training length 580 and Q=4.

Gamma: 0.00, MFV MSE: 0.006723
Gamma: 0.05, MFV MSE: 0.006723
Gamma: 0.10, MFV MSE: 0.006722
Gamma: 0.15, MFV MSE: 0.006722
Gamma: 0.20, MFV MSE: 0.006722
Gamma: 0.25, MFV MSE: 0.006722
Gamma: 0.30, MFV MSE: 0.006722
Gamma: 0.35, MFV MSE: 0.006723
Gamma: 0.40, MFV MSE: 0.006723
Gamma: 0.45, MFV MSE: 0.006724
Gamma: 0.50, MFV MSE: 0.006725
Gamma: 0.55, MFV MSE: 0.006726
Gamma: 0.60, MFV MSE: 0.006727
Gamma: 0.65, MFV MSE: 0.006728
Gamma: 0.70, MFV MSE: 0.006730
Gamma: 0.75, MFV MSE: 0.006731
Gamma: 0.80, MFV MSE: 0.006733
Gamma: 0.85, MFV MSE: 0.006735
Gamma: 0.90, MFV MSE: 0.006737
Gamma: 0.95, MFV MSE: 0.006740
Gamma: 1.00, MFV MSE: 0.006742

Best gamma: 0.20 (MFV MSE: 0.006722)

--- Skipping Bias-Variance Decomposition ---

Successfully completed run for NP_LL_Combo (without saving artifacts).

LocalPolynomialEngine will use device: cuda

Instantiated runner for NP_LL_Combo. Results will not be saved.

Structural break date identified: 2022-09-22

Full train/CV size: 6694, OOS test size: 40

--- (1/3) Tuning bandwidth for PRE-break model ---

Creating bandwidth grid with T=6115, d=1

Generated bandwidth grid: [0.01748722 0.02917044 0.04865922 0.08116847 0.13539717 0.2258561
0.37675068 0.62845801 1.04833114 1.74872173]

Set block size to 1223 for training length 6115 and Q=4.

Best bandwidth for PRE-break model: 1.7487

--- (2/3) Tuning bandwidth for POST-break model ---

Creating bandwidth grid with T=579, d=1

Generated bandwidth grid: [0.02801975 0.04673976 0.07796662 0.13005617 0.21694676 0.36188901
0.60366726 1.00697768 1.67974 2.8019752]

Set block size to 115 for training length 579 and Q=4.

Best bandwidth for POST-break model: 2.8020

--- (3/3) Tuning gamma for Convex Combination model ---

Set block size to 115 for training length 579 and Q=4.

Gamma: 0.00, MFV MSE: 0.007149
Gamma: 0.05, MFV MSE: 0.007110
Gamma: 0.10, MFV MSE: 0.007073
Gamma: 0.15, MFV MSE: 0.007038
Gamma: 0.20, MFV MSE: 0.007006
Gamma: 0.25, MFV MSE: 0.006976
Gamma: 0.30, MFV MSE: 0.006949
Gamma: 0.35, MFV MSE: 0.006923
Gamma: 0.40, MFV MSE: 0.006900
Gamma: 0.45, MFV MSE: 0.006879
Gamma: 0.50, MFV MSE: 0.006861
Gamma: 0.55, MFV MSE: 0.006844
Gamma: 0.60, MFV MSE: 0.006830
Gamma: 0.65, MFV MSE: 0.006819
Gamma: 0.70, MFV MSE: 0.006809
Gamma: 0.75, MFV MSE: 0.006802
Gamma: 0.80, MFV MSE: 0.006797
Gamma: 0.85, MFV MSE: 0.006794

Gamma: 0.90, MFV MSE: 0.006794
Gamma: 0.95, MFV MSE: 0.006796
Gamma: 1.00, MFV MSE: 0.006800
Best gamma: 0.90 (MFV MSE: 0.006794)

--- Skipping Bias-Variance Decomposition ---
Successfully completed run for NP_LL_Combo (without saving artifacts).
LocalPolynomialEngine will use device: cuda
Instantiated runner for NP_LL_Combo. Results will not be saved.
Structural break date identified: 2022-09-23
Full train/CV size: 6694, OOS test size: 40

--- (1/3) Tuning bandwidth for PRE-break model ---
Creating bandwidth grid with T=6116, d=1
Generated bandwidth grid: [0.01748665 0.02916948 0.04865763 0.08116582 0.13539275 0.22584871
0.37673836 0.62843745 1.04829685 1.74866455]
Set block size to 1223 for training length 6116 and Q=4.
Best bandwidth for PRE-break model: 1.7487

--- (2/3) Tuning bandwidth for POST-break model ---
Creating bandwidth grid with T=578, d=1
Generated bandwidth grid: [0.02802944 0.04675593 0.07799358 0.13010114 0.21702178 0.36201415
0.60387599 1.00732587 1.68032083 2.80294407]
Set block size to 115 for training length 578 and Q=4.
Best bandwidth for POST-break model: 2.8029

--- (3/3) Tuning gamma for Convex Combination model ---
Set block size to 115 for training length 578 and Q=4.

Gamma: 0.00, MFV MSE: 0.007153
Gamma: 0.05, MFV MSE: 0.007114
Gamma: 0.10, MFV MSE: 0.007076
Gamma: 0.15, MFV MSE: 0.007041
Gamma: 0.20, MFV MSE: 0.007008
Gamma: 0.25, MFV MSE: 0.006978
Gamma: 0.30, MFV MSE: 0.006949
Gamma: 0.35, MFV MSE: 0.006924
Gamma: 0.40, MFV MSE: 0.006900
Gamma: 0.45, MFV MSE: 0.006879
Gamma: 0.50, MFV MSE: 0.006860
Gamma: 0.55, MFV MSE: 0.006844
Gamma: 0.60, MFV MSE: 0.006829
Gamma: 0.65, MFV MSE: 0.006818
Gamma: 0.70, MFV MSE: 0.006808
Gamma: 0.75, MFV MSE: 0.006801
Gamma: 0.80, MFV MSE: 0.006796
Gamma: 0.85, MFV MSE: 0.006794
Gamma: 0.90, MFV MSE: 0.006793
Gamma: 0.95, MFV MSE: 0.006796
Gamma: 1.00, MFV MSE: 0.006800
Best gamma: 0.90 (MFV MSE: 0.006793)

--- Skipping Bias-Variance Decomposition ---
Successfully completed run for NP_LL_Combo (without saving artifacts).
LocalPolynomialEngine will use device: cuda
Instantiated runner for NP_LL_Combo. Results will not be saved.
Structural break date identified: 2022-09-26
Full train/CV size: 6694, OOS test size: 40

--- (1/3) Tuning bandwidth for PRE-break model ---
Creating bandwidth grid with T=6117, d=1
Generated bandwidth grid: [0.01748607 0.02916853 0.04865604 0.08116316 0.13538832 0.22584133
0.37672604 0.62841691 1.04826258 1.74860737]
Set block size to 1223 for training length 6117 and Q=4.
Best bandwidth for PRE-break model: 1.7486

--- (2/3) Tuning bandwidth for POST-break model ---

Creating bandwidth grid with T=577, d=1
Generated bandwidth grid: [0.02803915 0.04677212 0.0780206 0.1301462 0.21709695 0.36213954
0.60408517 1.00767479 1.68090286 2.80391496]
Set block size to 115 for training length 577 and Q=4.
Best bandwidth for POST-break model: 2.8039

--- (3/3) Tuning gamma for Convex Combination model ---

Set block size to 115 for training length 577 and Q=4.

Gamma: 0.00, MFV MSE: 0.007158

Gamma: 0.05, MFV MSE: 0.007119

Gamma: 0.10, MFV MSE: 0.007081

Gamma: 0.15, MFV MSE: 0.007046

Gamma: 0.20, MFV MSE: 0.007014

Gamma: 0.25, MFV MSE: 0.006983

Gamma: 0.30, MFV MSE: 0.006955

Gamma: 0.35, MFV MSE: 0.006929

Gamma: 0.40, MFV MSE: 0.006906

Gamma: 0.45, MFV MSE: 0.006885

Gamma: 0.50, MFV MSE: 0.006866

Gamma: 0.55, MFV MSE: 0.006849

Gamma: 0.60, MFV MSE: 0.006834

Gamma: 0.65, MFV MSE: 0.006822

Gamma: 0.70, MFV MSE: 0.006812

Gamma: 0.75, MFV MSE: 0.006805

Gamma: 0.80, MFV MSE: 0.006799

Gamma: 0.85, MFV MSE: 0.006796

Gamma: 0.90, MFV MSE: 0.006795

Gamma: 0.95, MFV MSE: 0.006797

Gamma: 1.00, MFV MSE: 0.006800

Best gamma: 0.90 (MFV MSE: 0.006795)

--- Skipping Bias-Variance Decomposition ---

Successfully completed run for NP_LL_Combo (without saving artifacts).

LocalPolynomialEngine will use device: cuda

Instantiated runner for NP_LL_Combo. Results will not be saved.

Structural break date identified: 2022-09-27

Full train/CV size: 6694, OOS test size: 40

--- (1/3) Tuning bandwidth for PRE-break model ---

Creating bandwidth grid with T=6118, d=1

Generated bandwidth grid: [0.0174855 0.02916758 0.04865445 0.08116051 0.13538389 0.22583394
0.37671372 0.62839636 1.04822831 1.7485502]

Set block size to 1223 for training length 6118 and Q=4.

Best bandwidth for PRE-break model: 1.7486

--- (2/3) Tuning bandwidth for POST-break model ---

Creating bandwidth grid with T=576, d=1

Generated bandwidth grid: [0.02804888 0.04678835 0.07804767 0.13019136 0.21717228 0.3622652
0.60429477 1.00802443 1.6814861 2.80488786]

Set block size to 115 for training length 576 and Q=4.

Best bandwidth for POST-break model: 2.8049

--- (3/3) Tuning gamma for Convex Combination model ---

Set block size to 115 for training length 576 and Q=4.

Gamma: 0.00, MFV MSE: 0.007161

Gamma: 0.05, MFV MSE: 0.007121

Gamma: 0.10, MFV MSE: 0.007084

Gamma: 0.15, MFV MSE: 0.007049

Gamma: 0.20, MFV MSE: 0.007015

Gamma: 0.25, MFV MSE: 0.006985

Gamma: 0.30, MFV MSE: 0.006956

Gamma: 0.35, MFV MSE: 0.006930

Gamma: 0.40, MFV MSE: 0.006906

Gamma: 0.45, MFV MSE: 0.006885

Gamma: 0.50, MFV MSE: 0.006866

Gamma: 0.55, MFV MSE: 0.006849

Gamma: 0.60, MFV MSE: 0.006834
Gamma: 0.65, MFV MSE: 0.006822
Gamma: 0.70, MFV MSE: 0.006812
Gamma: 0.75, MFV MSE: 0.006804
Gamma: 0.80, MFV MSE: 0.006799
Gamma: 0.85, MFV MSE: 0.006796
Gamma: 0.90, MFV MSE: 0.006795
Gamma: 0.95, MFV MSE: 0.006796
Gamma: 1.00, MFV MSE: 0.006800
Best gamma: 0.90 (MFV MSE: 0.006795)

--- Skipping Bias-Variance Decomposition ---
Successfully completed run for NP_LL_Combo (without saving artifacts).
LocalPolynomialEngine will use device: cuda
Instantiated runner for NP_LL_Combo. Results will not be saved.
Structural break date identified: 2022-09-28
Full train/CV size: 6694, OOS test size: 40

--- (1/3) Tuning bandwidth for PRE-break model ---
Creating bandwidth grid with T=6119, d=1
Generated bandwidth grid: [0.01748493 0.02916662 0.04865286 0.08115786 0.13537947 0.22582656
0.37670141 0.62837582 1.04819404 1.74849305]
Set block size to 1223 for training length 6119 and Q=4.
Best bandwidth for PRE-break model: 1.7485

--- (2/3) Tuning bandwidth for POST-break model ---
Creating bandwidth grid with T=575, d=1
Generated bandwidth grid: [0.02805863 0.04680461 0.0780748 0.13023661 0.21724777 0.36239112
0.60450481 1.00837481 1.68207056 2.8058628]
Set block size to 115 for training length 575 and Q=4.
Best bandwidth for POST-break model: 2.8059

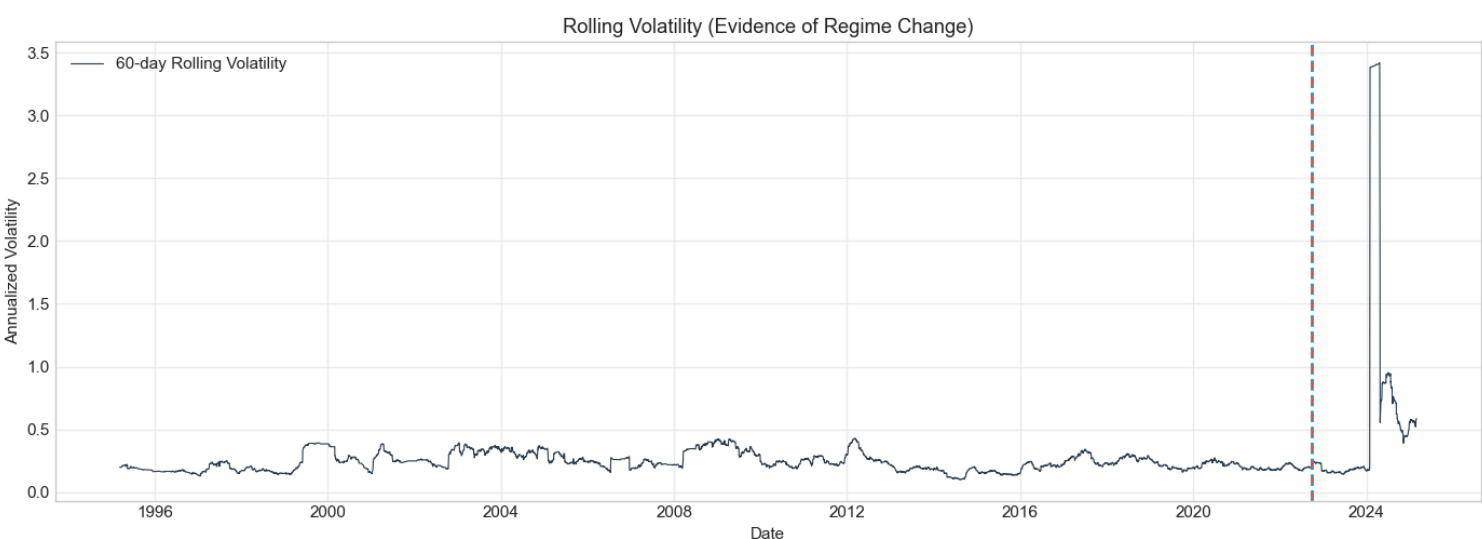
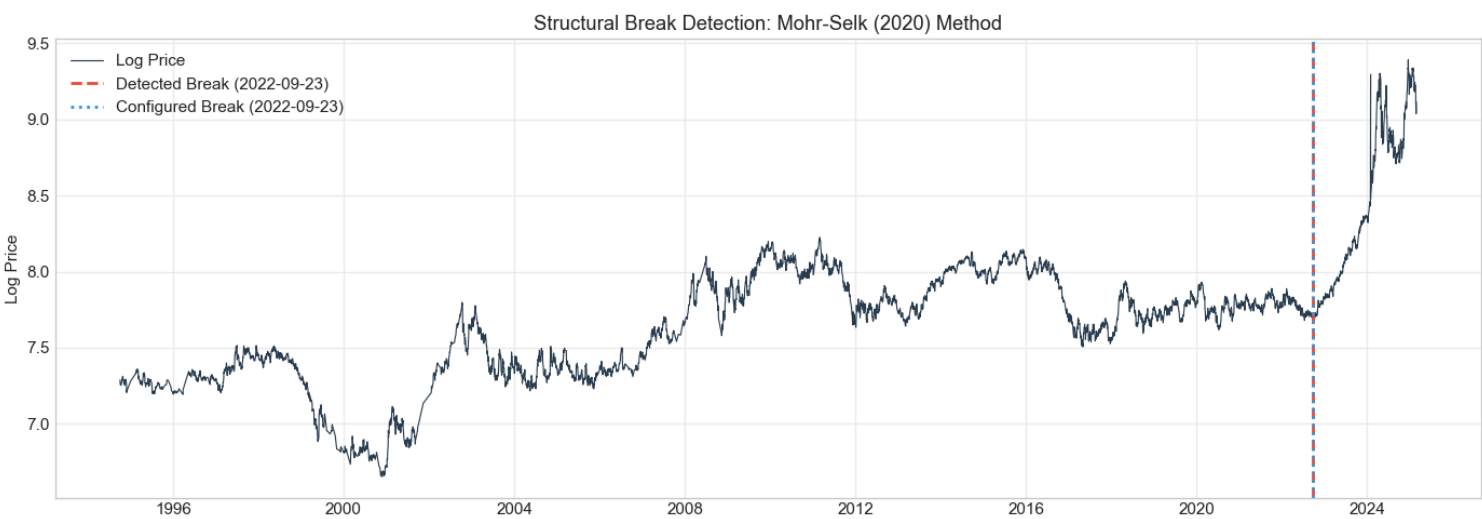
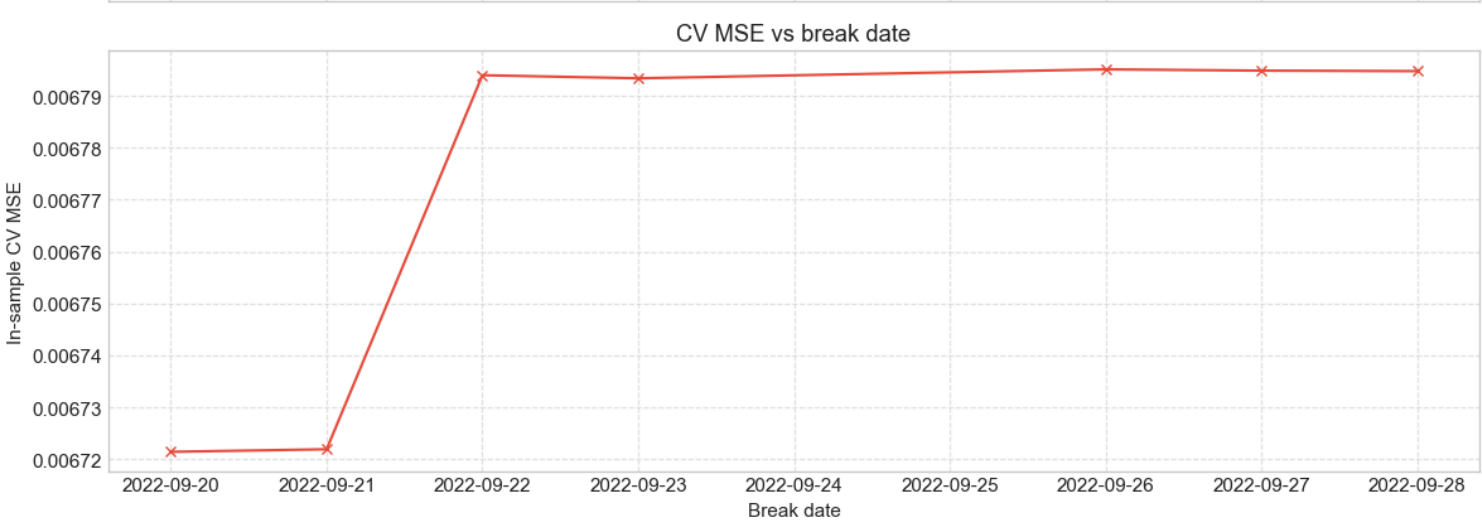
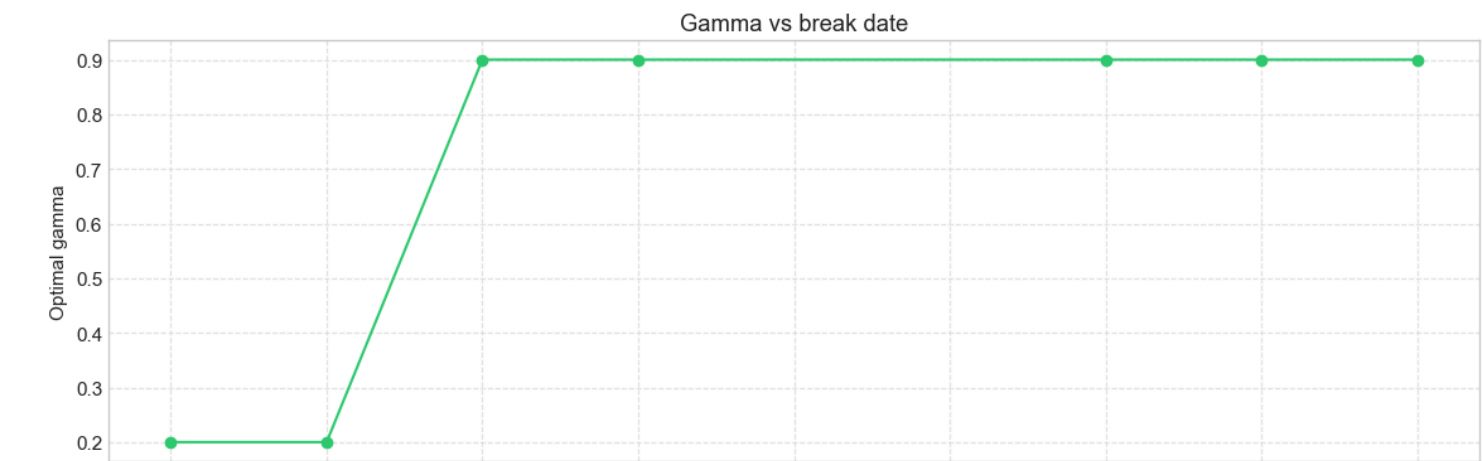
--- (3/3) Tuning gamma for Convex Combination model ---
Set block size to 115 for training length 575 and Q=4.

Gamma: 0.00, MFV MSE: 0.007167
Gamma: 0.05, MFV MSE: 0.007126
Gamma: 0.10, MFV MSE: 0.007088
Gamma: 0.15, MFV MSE: 0.007052
Gamma: 0.20, MFV MSE: 0.007019
Gamma: 0.25, MFV MSE: 0.006987
Gamma: 0.30, MFV MSE: 0.006959
Gamma: 0.35, MFV MSE: 0.006932
Gamma: 0.40, MFV MSE: 0.006908
Gamma: 0.45, MFV MSE: 0.006886
Gamma: 0.50, MFV MSE: 0.006867
Gamma: 0.55, MFV MSE: 0.006849
Gamma: 0.60, MFV MSE: 0.006835
Gamma: 0.65, MFV MSE: 0.006822
Gamma: 0.70, MFV MSE: 0.006812
Gamma: 0.75, MFV MSE: 0.006804
Gamma: 0.80, MFV MSE: 0.006799
Gamma: 0.85, MFV MSE: 0.006796
Gamma: 0.90, MFV MSE: 0.006795
Gamma: 0.95, MFV MSE: 0.006796
Gamma: 1.00, MFV MSE: 0.006800
Best gamma: 0.90 (MFV MSE: 0.006795)

--- Skipping Bias-Variance Decomposition ---
Successfully completed run for NP_LL_Combo (without saving artifacts).

	break_index	break_date	gamma	in_sample_cv_mse	oos_mse
0	6114	2022-09-20	0.2	0.006721	None
1	6115	2022-09-21	0.2	0.006722	None
2	6116	2022-09-22	0.9	0.006794	None
3	6117	2022-09-23	0.9	0.006793	None
4	6118	2022-09-26	0.9	0.006795	None

5 6119 2022-09-27 0.9 0.006795 None
6 6120 2022-09-28 0.9 0.006795 None



break index is 6117 break date is 2022-09-23

The Mohr–Selk detector flags a break at index 6117 (2022-09-23). Even by eye the big jump feels like it climaxes in 2024 (El Niño), but a 2022 start is still plausible: it captures the earlier shift in mean/vol driven by tightening supply conditions that lead into the 2024 spike. So the detected break is economically interpretable as the onset of the regime that later peaks with El Niño.

4. Experiment Configuration

```
Experiment Configuration:
Features: ['log_price_lagt']
Target: log_return_forecast_target
Training set: 6,694 observations
Test set (OOS): 40 observations
OOS Start: 2025-01-02
Break index (training): 6116
MFV Folds (Q): 4
```

5. Model Training

5.1 Non-Parametric Benchmarks

```
LocalPolynomialEngine will use device: cuda
Creating bandwidth grid with T=6694, d=1
Generated bandwidth grid: [0.01717366 0.02864739 0.04778673 0.07971307 0.13296941 0.22180635
0.36999529 0.61718934 1.02953386 1.71736599]
```

```
NP Configuration:
Kernel: Epanechnikov
Local Engine: Local Linear (order=1)
Dimension: 1
Bandwidth grid: 10 candidates
Range: [0.0172, 1.7174]
```

```
Training Pre-Break Local Linear Model...
Training size: 6,116
Selecting bandwidth via MFV...
Set block size to 1223 for training length 6116 and Q=4.
Selected bandwidth: 1.717366
MFV MSE: 0.00025423
Model trained.
```

```
Training Post-Break Local Linear Model...
Training size: 578
Selecting bandwidth via MFV...
Set block size to 115 for training length 578 and Q=4.
Selected bandwidth: 1.717366
MFV MSE: 0.00767773
Model trained.
```

```
Training Weighted Local Linear (WLL) Model...
Tuning gamma via MFV...
Gamma grid: 21 candidates
Set block size to 115 for training length 578 and Q=4.
Optimal gamma: 0.9500
Interpretation: 95.0% weight on pre-break model
MFV MSE: 0.00680709
Model trained.
```

5.2 Machine Learning Competitors

```
Training Random Forest...
  Grid size: 12 combinations
  Running MFV hyperparameter tuning...
Set block size to 1338 for training length 6694 and Q=4.
  Best params: {'n_estimators': 500, 'max_depth': 5, 'min_samples_leaf': 20, 'max_features': 'sqrt'}
  MFV MSE: 0.00082170
  Model trained.

Training XGBoost...
  Grid size: 32 combinations
  Running MFV hyperparameter tuning...
Set block size to 1338 for training length 6694 and Q=4.
  Best params: {'n_estimators': 200, 'max_depth': 3, 'learning_rate': 0.05, 'subsample': 1.0, 'colsample_bytree': 0.7}
  MFV MSE: 0.00082941
  Model trained.
```

5.3 ML Extension: Weighted ML Combinations

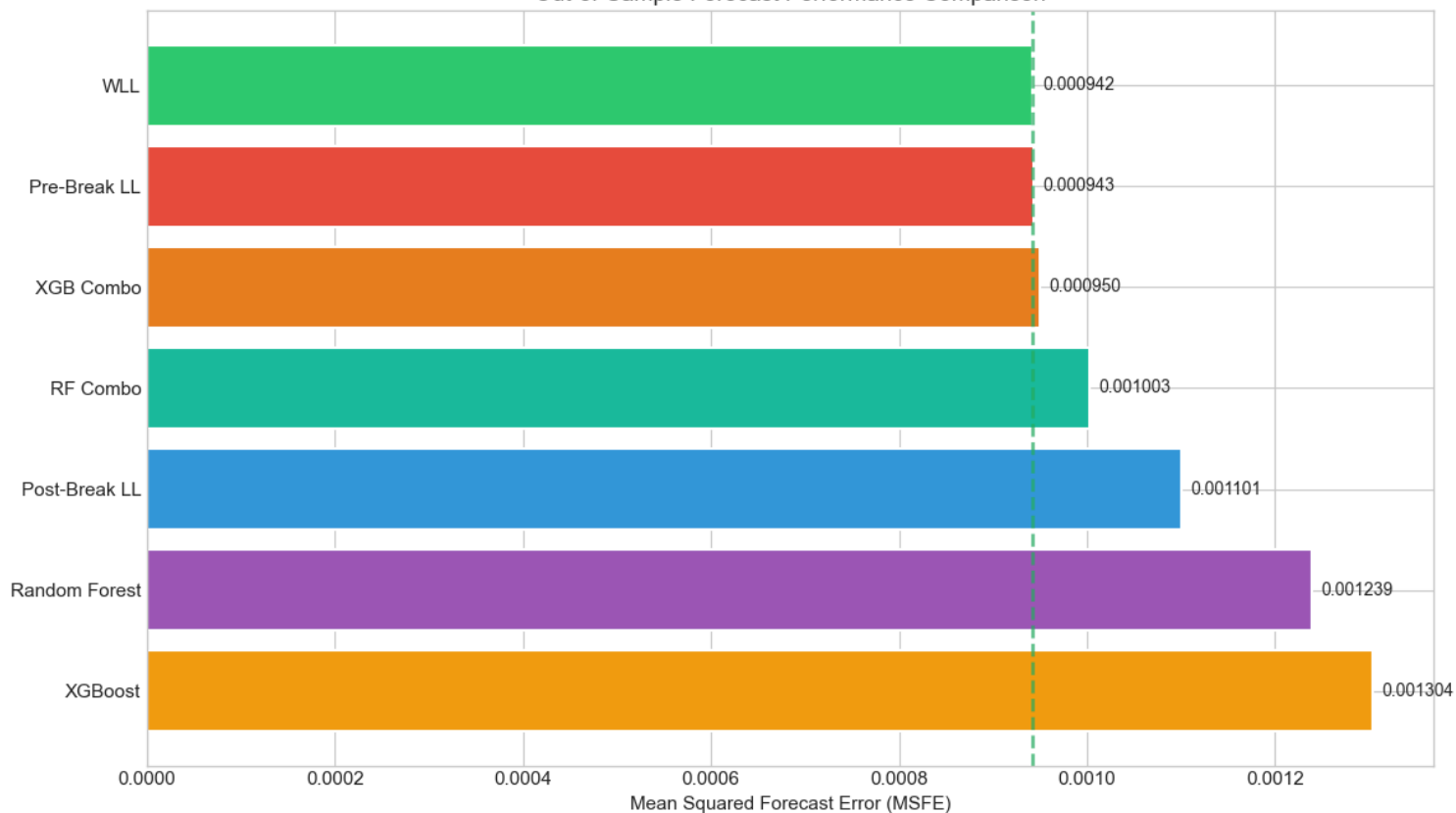
```
Training Weighted RF Combo...
Set block size to 115 for training length 578 and Q=4.
  Optimal gamma: 0.7000
Training Weighted XGB Combo...
Set block size to 115 for training length 578 and Q=4.
  Optimal gamma: 1.0000
All models trained.
```

6. Out-of-Sample Evaluation

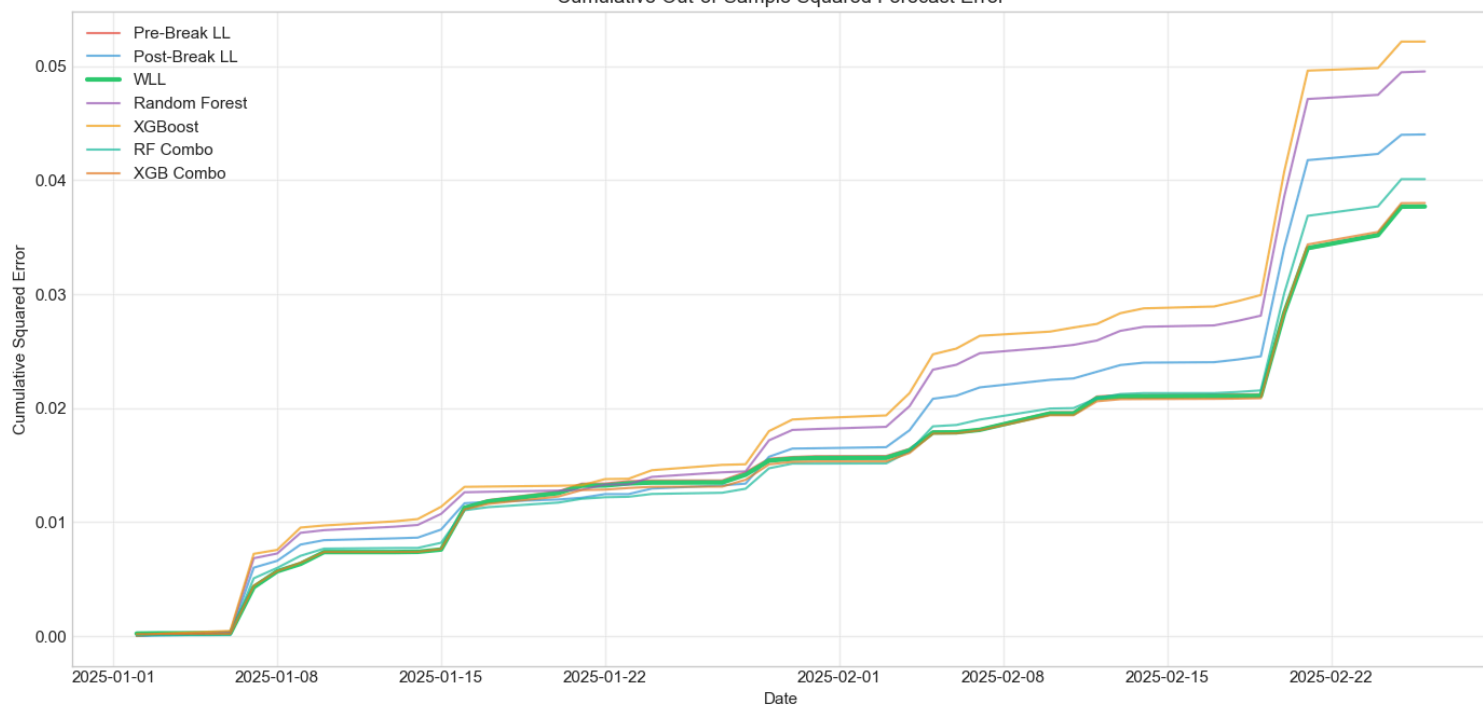
Generating out-of-sample predictions...

=====				
MEAN SQUARED FORECAST ERROR (MSFE) RESULTS				
=====				
Rank	Model	MSFE	RMSFE	Relative (%)
1	WLL	0.000942	0.030700	0.000000
2	Pre-Break LL	0.000943	0.030706	0.042678
3	XGB Combo	0.000950	0.030824	0.808613
4	RF Combo	0.001003	0.031663	6.371883
5	Post-Break LL	0.001101	0.033176	16.778409
6	Random Forest	0.001239	0.035196	31.433508
7	XGBoost	0.001304	0.036115	38.387813
=====				
Best Model: WLL (MSFE = 0.00094248)				

Out-of-Sample Forecast Performance Comparison



Cumulative Out-of-Sample Squared Forecast Error



7. Modified Diebold-Mariano Test

MDM Test Statistics (positive = row model worse than column):

	Pre-Break LL	Post-Break LL	WLL	Random Forest	XGBoost	RF Combo	XGB Combo
Pre-Break LL	0.000	-1.218	0.062	-1.733	-1.930	-0.807	-0.336
Post-Break LL	1.218	0.000	1.285	-3.218	-3.324	1.715	1.387
WLL	-0.062	-1.285	0.000	-1.804	-2.000	-0.889	-0.503
Random Forest	1.733	3.218	1.804	0.000	-3.396	2.435	1.932
XGBoost	1.930	3.324	2.000	3.396	0.000	2.659	2.134
RF Combo	0.807	-1.715	0.889	-2.435	-2.659	0.000	0.998
XGB Combo	0.336	-1.387	0.503	-1.932	-2.134	-0.998	0.000

P-Values:

	Pre-Break LL	Post-Break LL	WLL	Random Forest	XGBoost	RF Combo	XGB Combo
Pre-Break LL	1.000	0.231	0.951	0.091	0.061	0.425	0.738
Post-Break LL	0.231	1.000	0.206	0.003	0.002	0.094	0.173
WLL	0.951	0.206	1.000	0.079	0.052	0.379	0.618
Random Forest	0.091	0.003	0.079	1.000	0.002	0.020	0.061
XGBoost	0.061	0.002	0.052	0.002	1.000	0.011	0.039
RF Combo	0.425	0.094	0.379	0.020	0.011	1.000	0.324
XGB Combo	0.738	0.173	0.618	0.061	0.039	0.324	1.000

8. Summary

=====

EXPERIMENT SUMMARY

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Data:

Training: 1994-10-04 to 2025-01-02

Test: 2025-01-02 to 2025-02-26

Structural break: 2022-09-23

Hyperparameters:

Pre-break bandwidth: 1.717366

Post-break bandwidth: 1.717366

WLL gamma: 0.9500

RF Combo gamma: 0.7000

XGB Combo gamma: 1.0000

Results (MSFE):

- 1. WLL: 0.00094248 (0.00%)
- 2. Pre-Break LL: 0.00094288 (0.04%)
- 3. XGB Combo: 0.00095010 (0.81%)
- 4. RF Combo: 0.00100254 (6.37%)
- 5. Post-Break LL: 0.00110062 (16.78%)
- 6. Random Forest: 0.00123874 (31.43%)
- 7. XGBoost: 0.00130428 (38.39%)

Key Comparisons:

WLL vs Post-Break LL: -14.37%

WLL vs Random Forest: -23.92%

=====

Results saved to: w:\Research\NP\Cocoa\output\experiment_results

Hypotheses and how we test them

Main question: does explicit break-aware weighting (WLL) adapt faster and give better post-break forecasts than break-agnostic ML (RF, XGB)?

- H0 (overall): On the post-break test sample, WLL MSFE < RF and WLL MSFE < XGB.

- H1 (early): In the first K days of the post-break test, the WLL cumulative squared error grows slower than RF and XGB.

Data/timeline note: break detected at 2022-09-23; current OOS test starts 2025-01- 02 with 40 points. "Early" here means the first K test days. To make the adaptation

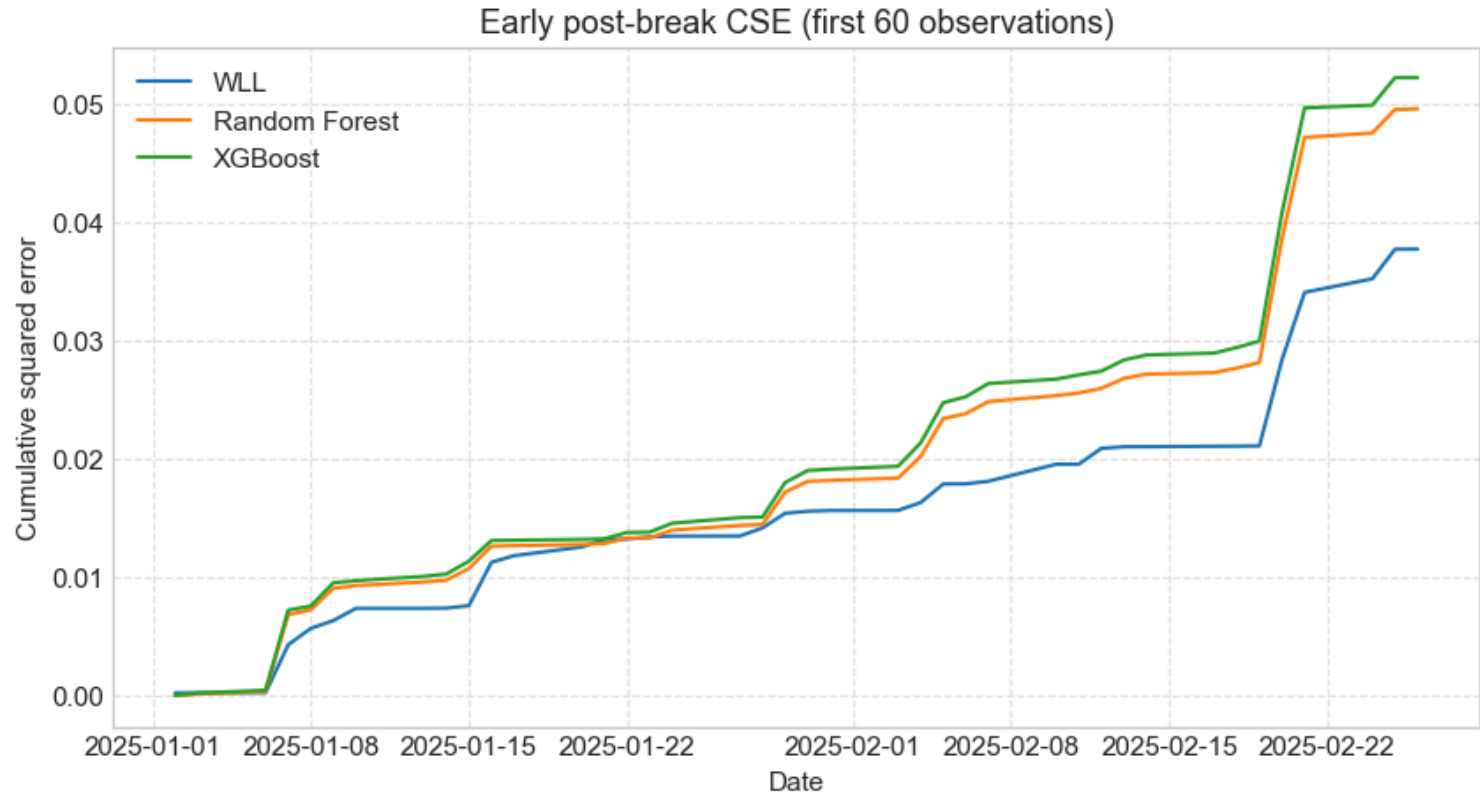
How we check:

- Use the saved `predictions_*.csv` from the main run.
- Compute overall MSFE for WLL, RF, XGB (H0).
- Compute and examine MSFE on short windows (1–20, 21–40) and the winning model in each state

Loaded predictions file: `predictions_20251201_205941.csv`

	Model	MSFE	Rel_to_WLL_%
0	WLL	0.000942	0.000000
1	Random Forest	0.001239	31.433508
2	XGBoost	0.001304	38.387814

	date	CSE_WLL	CSE_Random Forest	CSE_XGBoost
0	2025-01-02	0.000223	0.000004	0.000014
1	2025-01-03	0.000246	0.000152	0.000207
2	2025-01-06	0.000254	0.000353	0.000459
3	2025-01-07	0.004293	0.006832	0.007218
4	2025-01-08	0.005685	0.007247	0.007566



	Model	CSE_first_K	Rel_to_WLL_%
0	WLL	0.037699	0.000000
1	Random Forest	0.049550	31.433508
2	XGBoost	0.052171	38.387814

	Window	Model	MSFE	Rel_to_WLL_%
0	1-20	WLL	0.000770	0.000000
1	1-20	Random Forest	0.000859	11.587360
2	1-20	XGBoost	0.000899	16.836735
3	21-40	WLL	0.001115	0.000000
4	21-40	Random Forest	0.001619	45.131825
5	21-40	XGBoost	0.001709	53.262919

Conclusion 1–20 and 45–53% worse in 21–40). The early CSE curve stays below both ML curves. so its error grows slower (first-order “velocity”), not just lower total distance. That is a cleaner adaptation signal than the final CSE level alone.

DM p-values are weak because the test set is short, but point estimates favor WLL. Gamma: WLL ~0.95; RF Combo 0.7; XGB Combo 1.0. For a stronger “fast adaptation” the previous conclusion